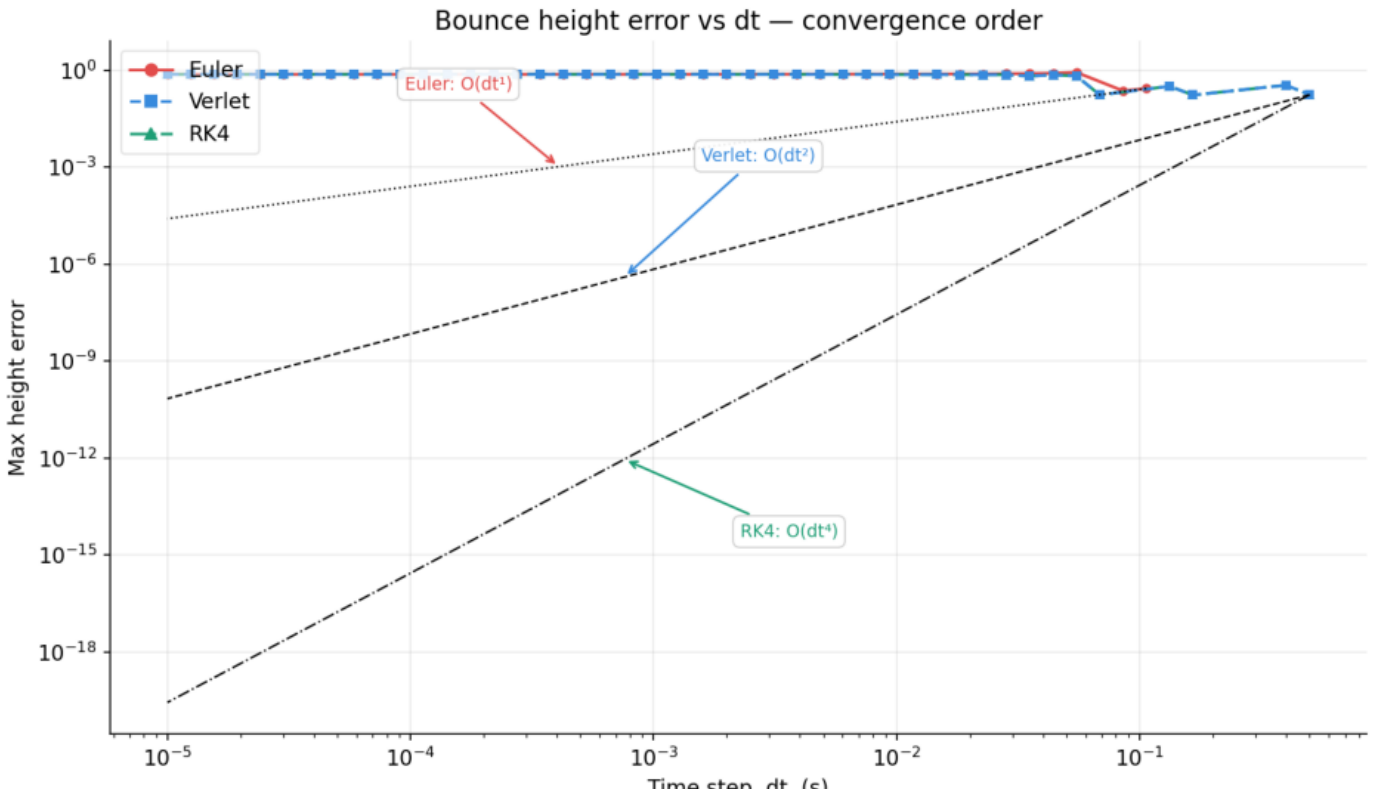
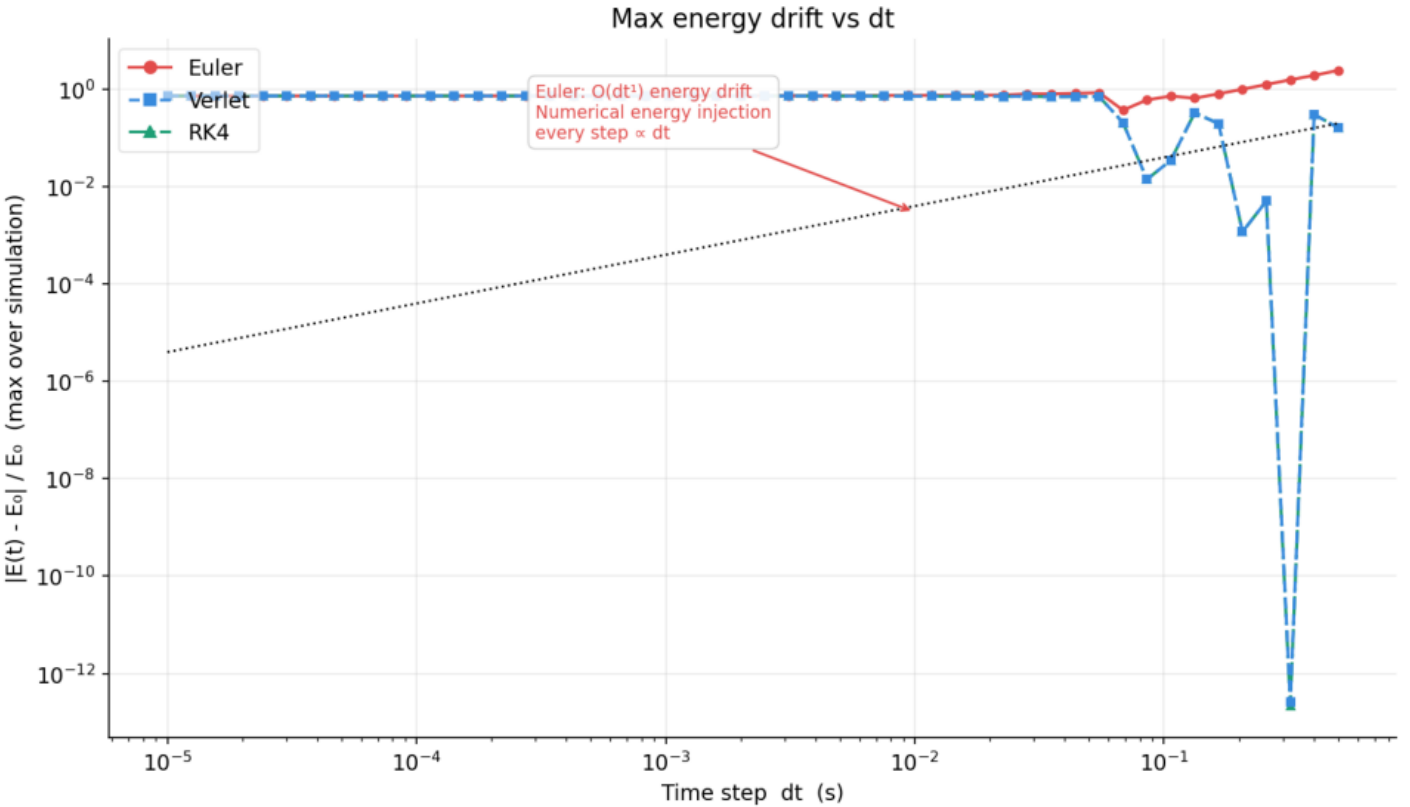
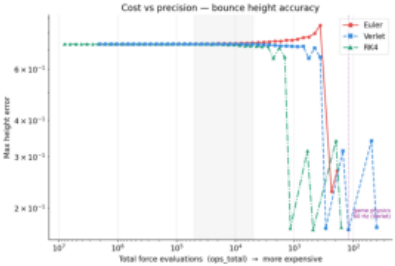


HEIGHT ERROR

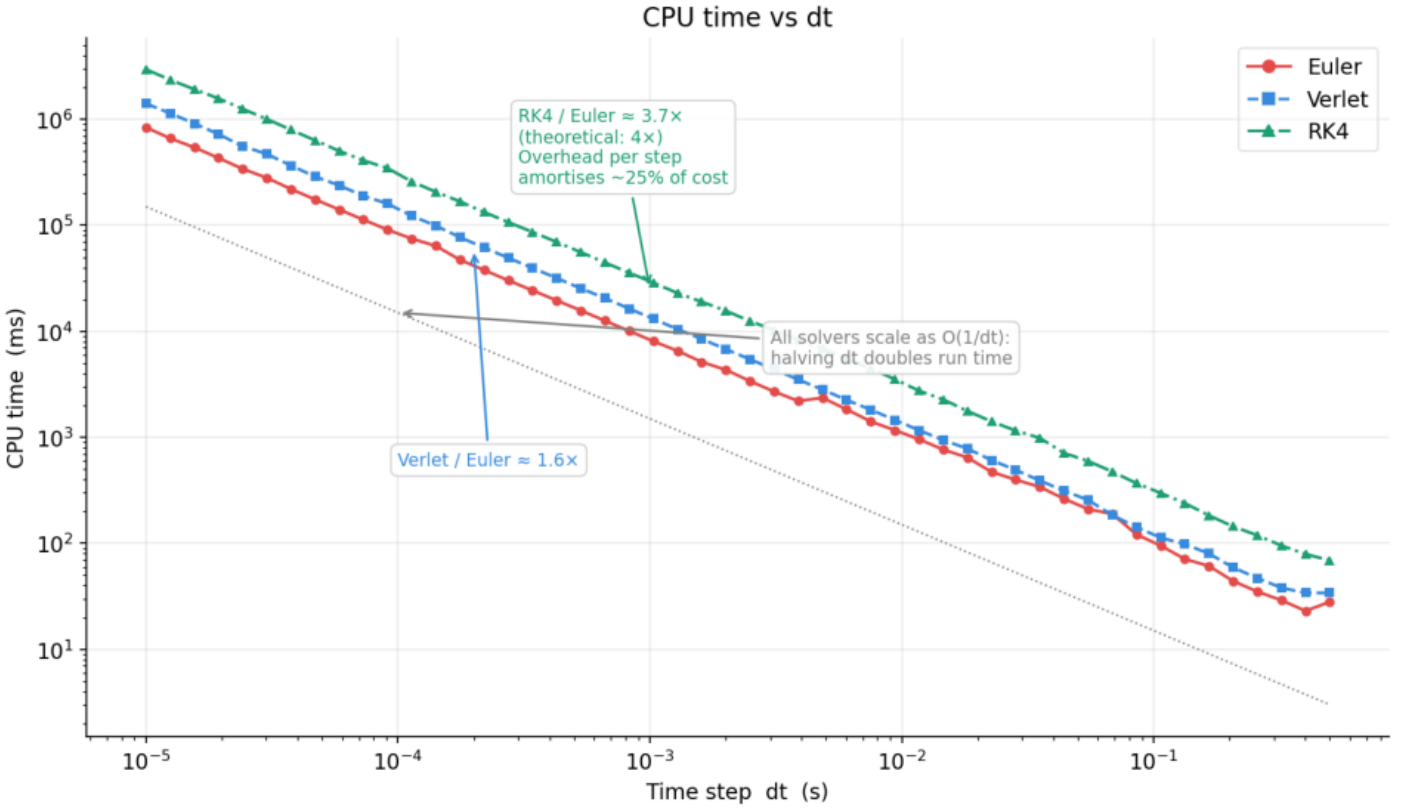
Measures deviation from analytical bounce decay:
$h_n = h_0 \cdot e^{2n}$



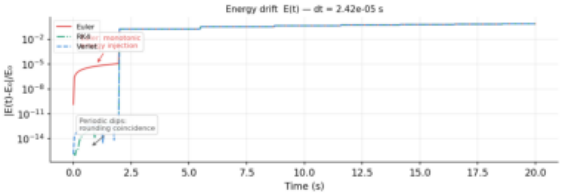
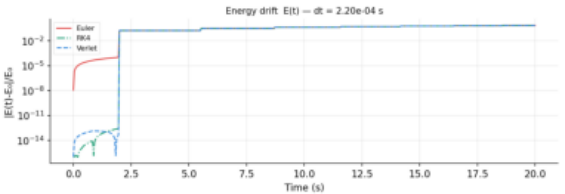
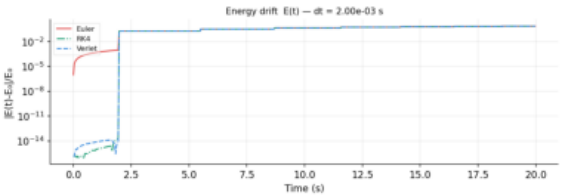
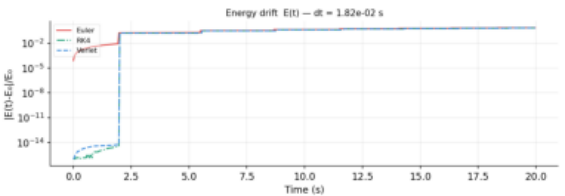
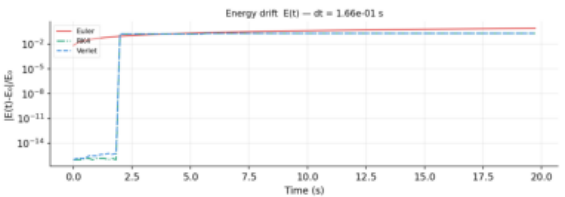
ENERGY CONSERVATION



COST vs PRECISION


CPU time vs dt
Euler
Verlet
RK4
RK4 / Euler ≈ 3.7×
(theoretical: 4×)
Overhead per step
amortises ~25% of cost
All solvers scale as O(1/dt):
halving dt doubles run time
Verlet / Euler ≈ 1.6×
CPU time (ms)
Time step dt (s)
10^{6}
10^{5}
10^{4}
10^{3}
10^{2}
10^{1}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
10^{-1}

CPU TIME



ENERGY DRIFT E(t)